UNIVERSITY FINAL PROJECT • MACHINE LEARNING

# Fake Job Postings Detector

A Machine Learning Approach to Screen Online Job Ads



Presenter Name · University Name
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# The Problem: Fake Job Postings

## Fake ads exploit urgency — and trust.

They waste applicants' time, expose personal information, and scale quickly across global job platforms.

**The need**

Automated screening that helps people and platforms review risk earlier.



**ONLINE JOB AD** → **ML RISK SCREEN** → **REAL / FLAGGED**

Classify each posting

Combine text + structured signals

Escalate suspicious ads for review

## A triage tool — not a final decision-maker

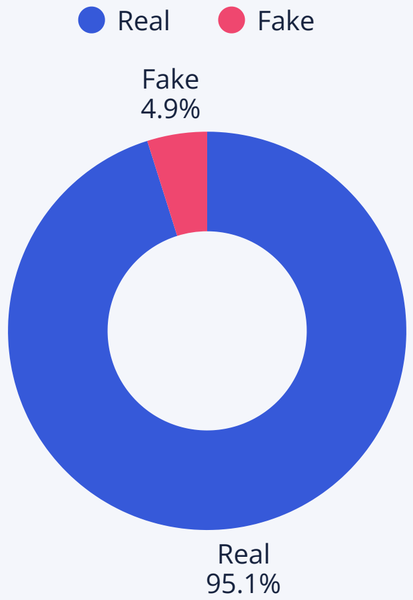
**17,599**
unique postings

**856**
fake postings

**18**
input columns

**TARGET ‘fraudulent’ • 1 = fake • 0 = real**

Class distribution




**Only 4.86% are fake**

# Two views of every job ad

### TEXT SIGNALS

Title
Description
Requirements
Benefits
Company profile

### STRUCTURED SIGNALS

Logo present
Screening questions
Contact information
Location
Employment type

# Four signals stood out

01

## Logos

Fake ads are less likely to show a company logo.

02

## Screening

Screening questions appear less often in fake ads.

03

## Contact

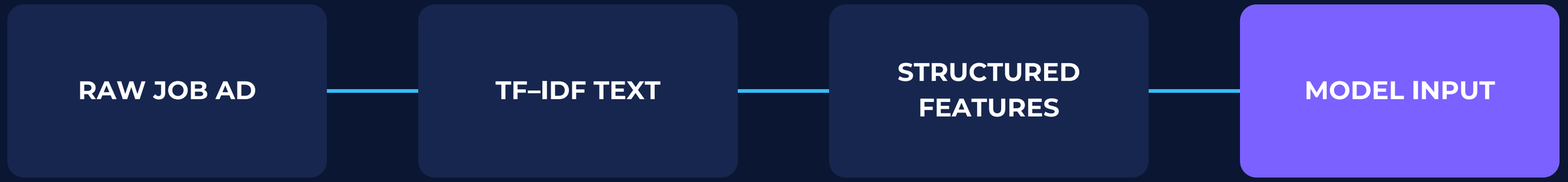
Email or phone details appear more often in fake ads.

04

## Repetition

Duplicates and near-duplicates reveal recurring campaigns.

# From raw advert to model-ready features




Words become weighted numeric signals

Logo • profile • contact • length • country

# Three models. One shared objective.

01

## LOGISTIC REGRESSION

Strong interpretable baseline

02

## LINEAR SVM

Excellent for sparse text

03

## RANDOM FOREST

Non-linear structured patterns

**Class weights counter the rarity of fake postings**

# Evaluate for the failure that matters

**GROUP-AWARE TRAIN SPLIT**

**HELD-OUT TEST SPLIT**

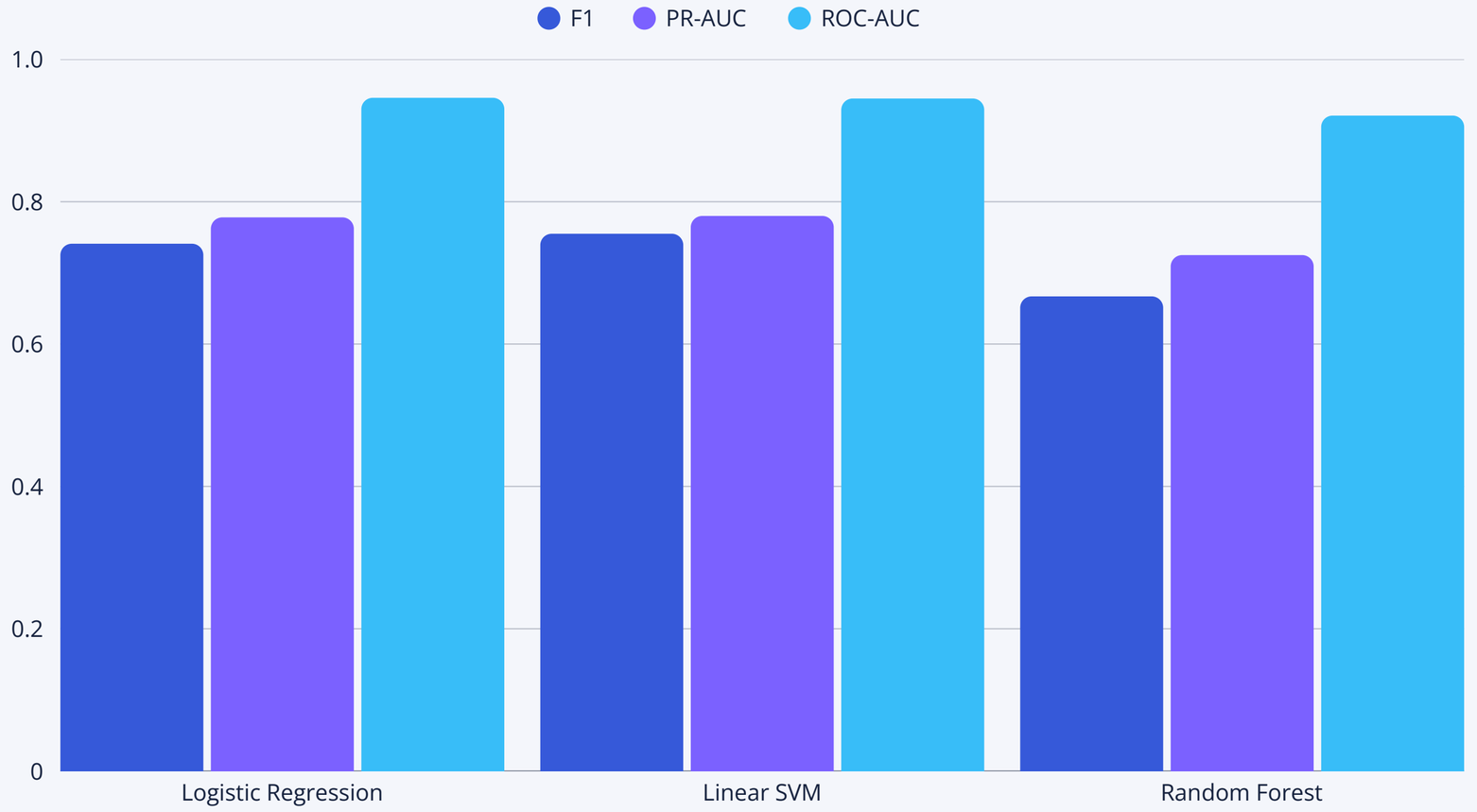
Near-duplicate ads stay together → less leakage

## PR–AUC

Best summary when fake ads are rare

PRECISION RECALL F1 ROC–AUC

# Comparing Model Performance




HIGHEST TEST F1

**0.755**

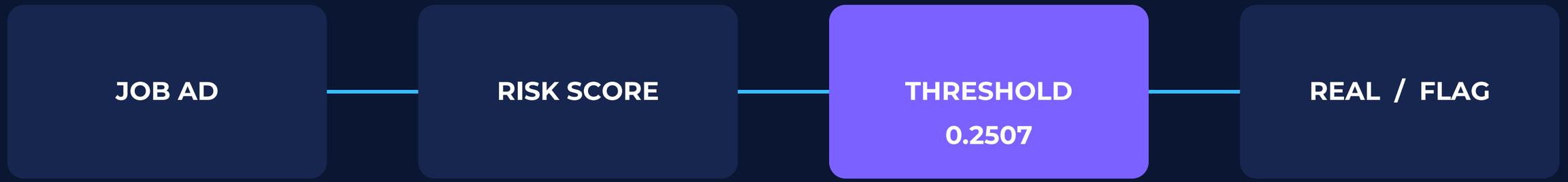
Linear SVM

**Final model selection was based on the predefined cross-validation procedure.**

**95.1% accuracy baseline**

Accuracy is misleading under severe class imbalance.

# A score becomes an actionable flag




Score ≥ 0.2507 → Flag for Review
Score < 0.2507 → Lower Risk

The cutoff is tuned with out-of-fold training predictions — never on the final test set.

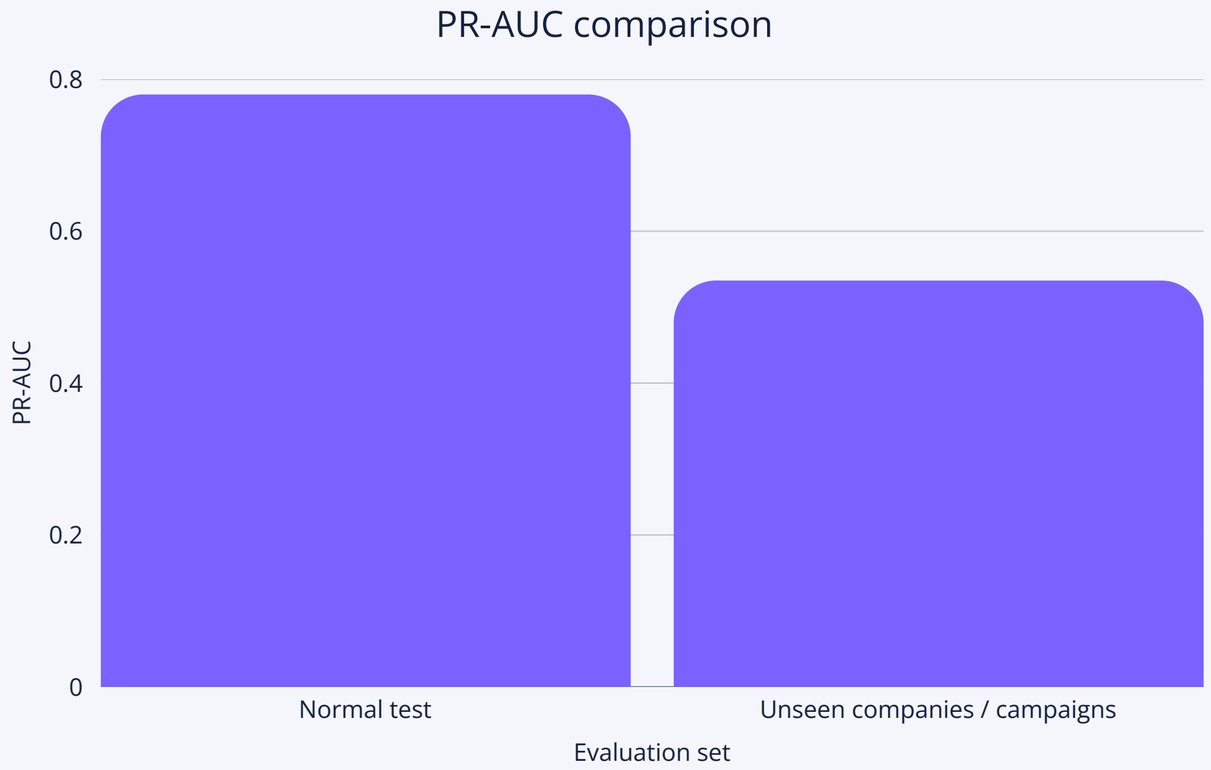
# The model combines weak signals

- **Missing company profile**
- **No logo**
- **No screening questions**
- **Direct contact details**
- **Recurring scam language**

## No single ‘magic’ word

Predictions emerge from patterns across content and metadata. Some patterns still reflect repeated templates or dataset-specific terms.

# Novel scams expose the generalization gap




**0.53–0.54**

PR–AUC on unseen scams

**Strong on known patterns. Weaker on completely new campaigns.**

**Use as triage — not certainty.**

FINAL APPLICATION

# Safer screening, clearer review.

A user-friendly web app turns the model into an accessible triage tool for job seekers and platforms.

**NEXT • Monitor drift • Refresh scam patterns • Keep humans in the loop**



## Thank you — Questions?